

Memory Augment Is All You Need for Image Restoration

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Abstract—The quality of images captured by consumer electronic devices, such as smartphones and digital cameras, is often compromised by adverse environmental conditions, leading to degradations like rain, shadows, blur, and low light. These issues significantly impact the user experience. This paper introduces MemoryNet, a novel image restoration framework designed to enhance the quality of images captured with consumer devices. We also propose Degradation-Aware CLIP (DA-CLIP) for the perceptual classification of degraded images, a common challenge in consumer photography. MemoryNet utilizes a three-granularity memory layer and contrastive learning to effectively restore images. The memory layer retains deep image features, while contrastive learning ensures the alignment of learned features for a balanced restoration. We have tested our model on several challenging tasks, including de-raining, de-shadowing, de-blurring, and low-light enhancement. The results show significant improvements in PSNR and SSIM in four datasets, demonstrating that MemoryNet can produce restored images with high perceptual authenticity, making it a promising solution to improve the imaging capabilities of consumer electronics. The code can be accessed in <https://github.com/zhangbaijin/MemoryNet>

Index Terms—Image restoration, memory augmentation, degradation-aware CLIP.

I. INTRODUCTION

IMAGE restoration pertains to a fundamental vision task focused on enhancing degraded images captured by consumer electronic devices. In the era of ubiquitous consumer electronics, smartphones, digital cameras, dashcams, and surveillance systems have become integral to daily life. However, images captured by these devices often suffer from various degradations, including noise, blur, low light conditions, and weather-related artifacts, significantly affecting the user experience and device functionality. The rapid progress in computer vision has enabled addressing an expanding array

of degradation challenges in consumer imaging applications, such as super-resolution for mobile photography, defogging for automotive cameras, de-shadowing for outdoor surveillance, removal of rain artifacts from dashcam footage, and enhancement of low-light images captured by smartphone. Image restoration is a fundamental yet inherently ill-posed inverse problem: a single degraded image can correspond to infinitely many plausible high-quality reconstructions, as real-world degradation processes—such as motion blur, rain streaks, shadows, or low-light noise—irreversibly discard visual information. In consumer electronics (e.g., smartphones, dashcams, surveillance cameras), this ambiguity is further amplified by the highly diverse and uncontrolled imaging conditions under which images are captured. Unlike controlled laboratory settings, consumer devices must handle arbitrary combinations of degradations without access to ground-truth degradation models or auxiliary inputs (e.g., shadow masks), making restoration both technically challenging and practically critical for user experience.

To regularize this ill-posed problem, image priors are essential to guide the solution toward perceptually plausible and structurally coherent results. Classical priors—such as sparsity or total variation—often fail to generalize across real-world scenes and device-specific artifacts. While modern deep learning approaches have made significant progress by learning implicit priors from large-scale data, most existing methods focus narrowly on architectural innovation (e.g., attention mechanisms, transformers, or Generative Adversarial Networks (GANs) [1], [2]). Crucially, they overlook a key insight from human vision: the ability to restore corrupted perception often relies on memory of typical structures and patterns—such as the expected shape of a car, the texture of pavement, or the lighting consistency of a scene. This structural memory enables robust inference even under severe degradation, yet it remains underexplored in current consumer-oriented restoration frameworks.

Moreover, practical deployment in consumer electronics imposes stringent constraints: algorithms must be lightweight, fast, and mask-free, while still delivering high perceptual quality. Unfortunately, many state-of-the-art methods either require auxiliary inputs (e.g., shadow masks [1], [3]), rely on heavy generative models unsuitable for on-device inference, or fail to leverage semantic guidance in a degradation-aware manner. Meanwhile, recent advances in vision-language models (e.g., Contrastive Language-Image Pre-training (CLIP))

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offer powerful semantic priors but are ill-suited for low-level restoration tasks due to their insensitivity to fine-grained structural details and their assumption of clean inputs.

To address these gaps, we propose MemoryNet, a novel image restoration framework that explicitly integrates hierarchical structural memory with degradation-aware semantic guidance—tailored for real-world consumer electronics. Our approach is motivated by two key observations: (1) Memory-augmented representation enables the model to recall and reconstruct prototypical patterns at multiple granularities (part, instance, and semantic levels), thereby constraining the solution space toward structurally consistent outputs without requiring masks or paired degradation labels; (2) Pre-trained vision-language models can be adapted for low-level tasks if made aware of input degradation, allowing them to provide meaningful perceptual supervision even when the input is severely corrupted.

Accordingly, we introduce two core components: (i) a three-granularity memory augmentation module that encodes learnable structural prototypes to enhance restoration fidelity and generalization; and (ii) a Degradation-Aware CLIP (DA-CLIP) that repurposes CLIP’s semantic knowledge for perceptual classification and guidance under degradation.

MemoryNet is designed to be lightweight, mask-free, and end-to-end trainable, making it suitable for real-time deployment on consumer devices. It achieves strong performance across four representative tasks—shadow removal, raindrop cleaning, motion deblurring, and low-light enhancement—demonstrating that memory is indeed “all you need” to bridge the gap between real-world degradation and perceptually authentic restoration.

The key contributions of this paper are as follows:

- 1) We developed an innovative memory augmentation module that models a learnable latent property variable to capture globally representative structural prototype patterns, optimized for efficient execution on consumer device processors.
- 2) We proposed a degradation-aware visual language model (DA-CLIP) to effectively adapt the pre-trained visual language model for low-level visual tasks in consumer electronics, serving as a comprehensive framework for image restoration in smartphones, cameras, and other consumer imaging devices.
- 3) We conducted extensive experiments across four common image restoration tasks relevant to consumer electronics: synthetic image de-shadowing for outdoor photography, real image deraining for dashcam applications, image deblurring for handheld camera shake correction, and image low-light enhancement for smartphone night photography.

II. RELATED WORK

A. The Development of Memory Modules

He et al. [4] pioneered the use of a memory module for anomaly detection, conceptualizing the Encoder as a means to generate queries. The Decoder employs fresh feature maps from the Memory module to reconstruct images, thereby

accentuating normal frames while magnifying reconstruction errors in anomalous frames. Park et al. [5] advances the Auto Encoder with information on normal frames to enhance the distinction between normal and abnormal frames in video anomaly detection. For de-raining tasks, MMOS [6] integrates an intermediate Memory module to model and store diverse rain patterns, assisting in the rain removal process.

B. Vision-Language Models

Recent research underscores the promising potential of using pre-trained Vision-Language Models (VLMs) to improve subsequent tasks by leveraging universal visual and text representations [7], [8], [9]. Typically, a VLM consists of a text encoder and an image encoder, which work to obtain aligned multimodal features from noisy image-text pairs through contrastive learning. The BLIP approach refines this by filtering out noisy data from the web using generated captions. While VLMs excel in zero-shot and label-free classification for downstream tasks, their success in image restoration has been limited due to the necessity for precise, specialized vocabulary. A novel technique for refining vision-language models is known as prompt learning [10], where words within the prompt context are represented by learnable vectors which are then optimized for the specific task. The degradation-aware visual language model DA-CLIP [11] incorporates an Image Controller to predict degradation types and uses a CLIP Image Encoder module to extract high-quality content embeddings, integrating DA-CLIP with a reconstruction network. For the training of DA-CLIP, a hybrid dataset featuring 10 types of degradation is constructed.

III. NETWORK STRUCTURE

The MemoryNet architecture is depicted in Fig. 1. This paper presents the Degradation-aware CLIP (DA-CLIP) [11], designed for low-level visual applications, offering a uniform approach to image restoration. DA-CLIP creatively incorporates an extra controller to alter the standard CLIP image encoder, which aids in forecasting enhanced feature embeddings. This process allows for the development of adaptive classifiers tailored to diverse degradation scenarios such as blur, dim lighting, shadows, and rain.

A. Memory Augmentation

1) *How to Detect Abnormal Area?*: Blur is categorized as a type of irregular pattern. To distinguish and convert it into a normal pattern, we employ a mixture of abnormal detection and a completion proxy. This method aligns well with the transformer framework. Within this framework, shadow images are treated as abnormal, whereas clean images are deemed normal.

Our anomaly detection model adopts the conventional Encoder-Decoder architecture. Within this framework, pristine images are processed by a memory-augmented encoder/decoder to capture standard patterns. This technique draws on the approach described in [12]. It is not necessary to alter the fundamental encoder/decoder of your existing system. Just integrate memory and guidance into it. Once the

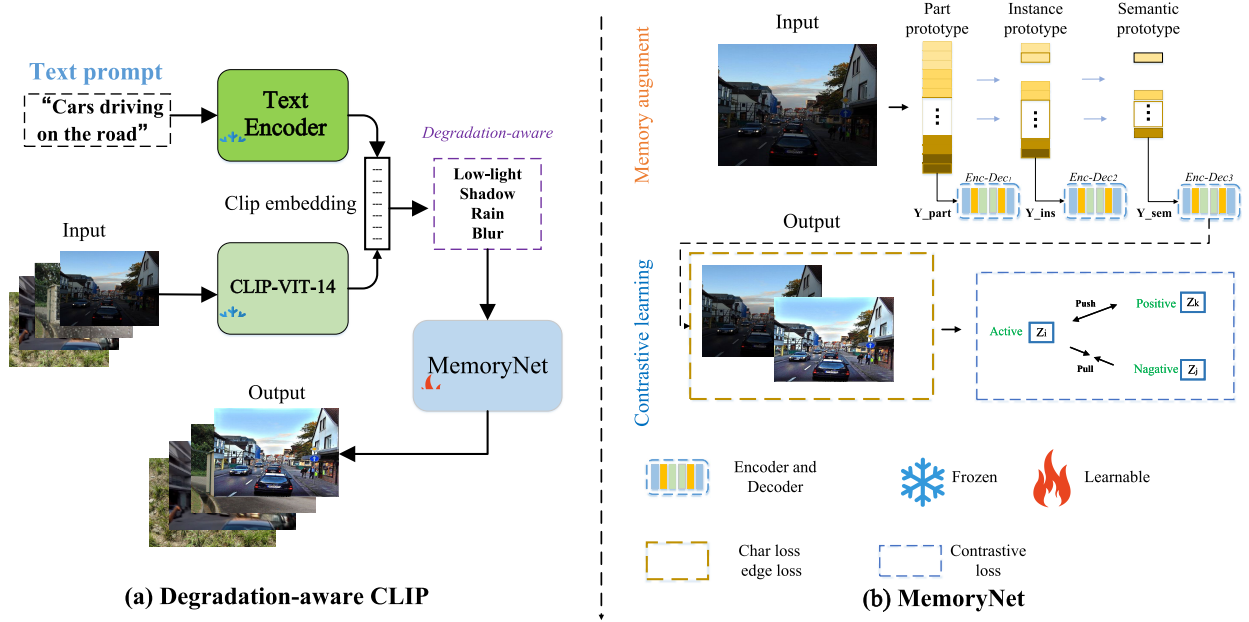


Fig. 1. Overview of the proposed MemoryNet framework. (a) The degradation-aware CLIP module identifies image degradation types through text-guided classification. (b) The core MemoryNet architecture features hierarchical memory augmentation and contrastive learning mechanisms for image restoration.

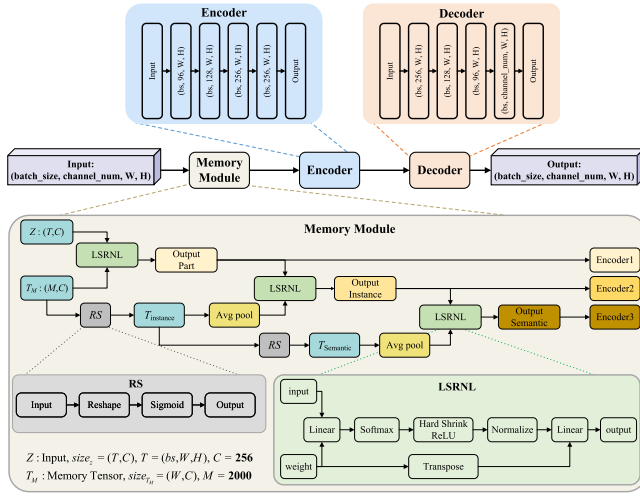


Fig. 2. Detailed illustration of the memory augmentation component. Input features are processed through an encoder yielding tensors of shape $(bs, 256, W, H)$, subsequently transformed to $(bs, W, H, 256)$ format. The memory mechanism employs a learnable matrix T_M $((M, C)$ dimensions) to compute attention weights through matrix multiplication. A hard threshold of 0.0025 is applied followed by normalization to generate the final output features.

anomaly detection training phase is completed, the encoder and decoder can reconstruct images, with the memory preserving normal patterns. Nevertheless, if a blurred image is input, the model fails to restore it to its original clarity.

2) *Memory Module*: The configuration of Memory augmentation is depicted in Fig. 2. When abnormal frames are processed by the CNN model, its high capacity can lead to low reconstruction errors and subsequently, incorrect results. To address this challenge, we have embedded a Memory Module in both the Encoder and Decoder. This inclusion

enables the capture of typical frame attributes and decreases the CNN's capacity to characterize them, thereby improving the differentiation between normal and abnormal frames. Furthermore, in our research, we have redesigned the interpretability of probabilities processed in the final CNN layer for classification-based ranking retrieval. We have also introduced a hierarchical memory adjustment and alignment module to mitigate domain bias.

The memory module consists of N prototypes, depicted by a matrix $\mathbf{M} \in \mathbb{R}^{N \times C}$, where C refers to the constant feature dimension. The parameter N , representing the number of memory items, can be adjusted. Addressing memory involves determining how relevant each query is to all memory items, which is then followed by the use of an attention-based addressing mechanism. This mechanism, known as the memory reader, serves to assign each image to a suitable prototype:

$$w_{ij} = \frac{\exp(d(\mathbf{f}_i, \mathbf{m}_j))}{\sum_{j=1}^N \exp(d(\mathbf{f}_i, \mathbf{m}_j))},$$

$$d(\mathbf{f}_i, \mathbf{m}_j) = \frac{\mathbf{f}_i \mathbf{m}_j^T}{\|\mathbf{f}_i\| \|\mathbf{m}_j\|}, \quad (1)$$

where $\mathbf{f}_i$ and $\mathbf{m}_j$ are the feature and prototype slice prototype matrix $\mathbf{M}$ from input $\mathbf{f}$. w_{ij} is the normalized weight to measure the cosine similarity $d(\cdot, \cdot)$ between $\mathbf{f}_i$ and $\mathbf{m}_j$. Therefore, the assigned prototype from feature $\mathbf{f}$ can be calculated h as:

$$\mathbf{y} = \text{Memory}(\mathbf{f}, \mathbf{M}) = \sum_{i=1}^{H \times W} \sum_{j=1}^N w_{ij} \mathbf{m}_j. \quad (2)$$

Let $\mathbf{f}_i \in \mathbb{R}^C$ denote the i -th spatial feature vector extracted from the input feature map $\mathbf{f} \in \mathbb{R}^{H \times W \times C}$, and let $\mathbf{m}_j \in \mathbb{R}^C$ denote the j -th prototype in the memory matrix $\mathbf{M} = [\mathbf{m}_1, \dots, \mathbf{m}_N]^T \in \mathbb{R}^{N \times C}$. The attention weight w_{ij} is

computed as the normalized cosine similarity between $\mathbf{f}_i$ and $\mathbf{m}_j$:

$$w_{ij} = \frac{\exp(d(\mathbf{f}_i, \mathbf{m}_j))}{\sum_{k=1}^N \exp(d(\mathbf{f}_i, \mathbf{m}_k))}, \quad d(\mathbf{f}_i, \mathbf{m}_j) = \frac{\mathbf{f}_i^\top \mathbf{m}_j}{\|\mathbf{f}_i\| \|\mathbf{m}_j\|}. \quad (3)$$

The memory-augmented output feature map $\mathbf{y} \in \mathbb{R}^{H \times W \times C}$ is then obtained by aggregating the prototypes according to the attention weights:

$$\mathbf{y} = \text{Memory}(\mathbf{f}, \mathbf{M}) = \sum_{i=1}^{H \times W} \sum_{j=1}^N w_{ij} \mathbf{m}_j. \quad (4)$$

The Memory Augmentation (MA) module is designed to capture hierarchical semantics through three granularities: part, instance, and semantic levels. The full memory matrix $\mathbf{M}$ has the structure $2 \times (\mathcal{P} \times \mathcal{I} \times \mathcal{S} \times N_c) \times C$, where $\mathcal{P}$, $\mathcal{I}$, and $\mathcal{S}$ denote the number of prototypes at the part, instance, and semantic levels, respectively, and N_c is the number of semantic categories. Each higher-level prototype is constructed as a weighted aggregation of lower-level prototypes. Specifically, the i -th instance-level prototype $\mathbf{m}_{\text{ins},i} \in \mathbb{R}^C$ is defined as:

$$\mathbf{m}_{\text{ins},i} = \frac{1}{\mathcal{P} \mathcal{S} N_c} \sum_{j=(\mathcal{P} \mathcal{S} N_c)(i-1)+1}^{(\mathcal{P} \mathcal{S} N_c)i} \alpha \mathbf{m}_{\text{part},j}, \quad (5)$$

where α is a learnable scalar weight (or can be absorbed into the prototype learning process).

The hierarchical memory readout is performed sequentially:

$$\begin{aligned} \mathbf{y}_{\text{part}} &= \text{Memory}(\mathbf{f}, \mathbf{M}_{\text{part}}), \\ \mathbf{y}_{\text{ins}} &= \text{Memory}(\mathbf{h}_{\text{part}}, \mathbf{M}_{\text{ins}}), \\ \mathbf{y}_{\text{sem}} &= \text{Memory}(\mathbf{h}_{\text{ins}}, \mathbf{M}_{\text{sem}}), \end{aligned} \quad (6)$$

where $\mathbf{h}_{\text{part}}$ and $\mathbf{h}_{\text{ins}}$ denote intermediate feature representations passed from the previous stage.

During the encoding phase, the memory-augmented features are fused with shallow features from the decoder via skip connections. The inputs to the three encoder stages are given by:

$$\begin{aligned} \text{Enc1}_{\text{input}} &= \mathbf{y}_{\text{part}}, \\ \text{Enc2}_{\text{input}} &= \mathbf{y}_{\text{ins}} + \mathbf{SFe}_{\text{Dec1}}, \\ \text{Enc3}_{\text{input}} &= \mathbf{y}_{\text{sem}} + \mathbf{SFe}_{\text{Dec2}}, \end{aligned} \quad (7)$$

where $\mathbf{SFe}_{\text{Deck}}$ denotes the shallow feature embedding from the k -th decoder block.

To ensure faithful reconstruction of normal patterns and effective suppression of anomalies, we impose a reconstruction constraint:

$$\mathcal{L}_{\text{recon}} = \|\text{Mem}(\text{Dec}(\text{Enc}(\mathbf{Y}))) - \mathbf{Y}\|_2, \quad (8)$$

where $\mathbf{Y}$ is the ground-truth clean image, and $\text{Enc}(\cdot)$, $\text{Dec}(\cdot)$, and $\text{Mem}(\cdot)$ denote the encoder, decoder, and memory readout operations, respectively.

B. Loss Function Design

The overall loss function $\mathcal{L}$ is a weighted sum of three components:

$$\mathcal{L} = \sum_{s=1}^3 [\mathcal{L}_{\text{char}}(\mathbf{X}_s, \mathbf{Y}) + \lambda \mathcal{L}_{\text{edge}}(\mathbf{X}_s, \mathbf{Y})] + \mathcal{L}_{\text{recon}}, \quad (9)$$

where $\mathbf{X}_s$ denotes the output at scale s , and $\mathbf{Y}$ is the ground-truth image. The term $\mathcal{L}_{\text{char}}$ is the Charbonnier loss, which provides robust pixel-wise supervision. The term $\mathcal{L}_{\text{edge}}$ is an edge loss that preserves high-frequency structural details. Finally, $\mathcal{L}_{\text{recon}}$ is the $\mathcal{L}_{\text{MSE}}$ reconstruction constraint that ensures the memory module learns representative patterns.

$$\mathcal{L}_{\text{char}} = \sqrt{\|\mathbf{X}_s - \mathbf{Y}\|^2 + \varepsilon^2}, \quad (10)$$

with the constant ε empirically set to 10^{-3} for all experiments. In addition, $\mathcal{L}_{\text{edge}}$ is the edge loss, defined as:

$$\mathcal{L}_{\text{edge}} = \sqrt{\|\Delta(\mathbf{X}_s) - \Delta(\mathbf{Y})\|^2 + \varepsilon^2} \quad (11)$$

IV. EXPERIMENT

A. Implementation Details

MemoryNet is a fully trainable model from scratch, eliminating the need for pre-training. It was developed with PyTorch 1.8.0 and an NVIDIA GTX 3090 GPU. Our evaluation in this study utilized metrics such as PSNR, SSIM, and RMSE. This study employs the ISTD dataset as referenced in [1], DeRainDrop dataset [28], LOL-v2 dataset [29] and GoPro dataset [30] for deblurring images. For optimization of our network, we utilize the Adam optimizer with $\beta_1 = 0.9$ and $\beta_2 = 0.999$. Learning Rate: An initial learning rate of 2×10^{-4} was used, which was gradually reduced using a cosine annealing scheduler over the training process. Batch Size: 16 for all experiments. Epochs: Training was conducted for 300 epochs on each dataset, with early stopping based on validation performance.

B. Comparison With State-of-the-Art Methods

1) *Shadow Removal*: Our approach is evaluated against a range of existing methods, such as Yang [13], Guo [14], Gong [15], DeShadowNet [16], STC-GAN [1], DSC [18], Mask-ShadowGAN [23], RIS-GAN [19], DHAN [20], SID [17], LG-shadow [24], G2R [22], DC-ShadowNet [53], Auto-exp [21], SpA-Former [25], Shadow-Refiner [26], ShadowFormer [3] and Diff-Shadow [27]. For assessment, we use the root mean square error (RMSE), structural similarity index (SSIM), and Peak Signal-to-Noise Ratio (PSNR) in the LAB color space. Table 1 details the RMSE, SSIM, and PSNR outcomes for these shadow removal techniques on the ISTD dataset [1]. The comprehensive evaluation is depicted in Fig. 3. MemoryNet excels in PSNR across partially, non-partially shaded, and unshaded regions. In unshaded areas, our RMSE also sets a new standard, outperforming the state-of-the-art generally. SID [17] and G2R [22] sometimes inaccurately assess dark non-shaded areas, leading to estimation errors. Their models do not fully leverage the shadow mask data,

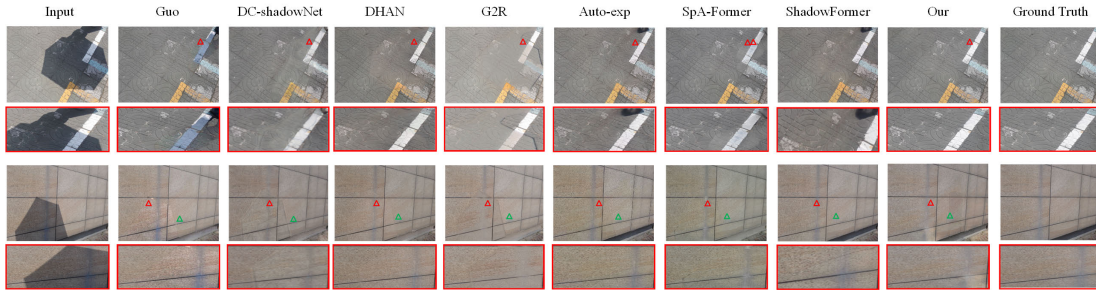


Fig. 3. Qualitative results demonstrating shadow removal performance on the ISTD benchmark dataset.

TABLE I
QUANTITATIVE EVALUATION OF SHADOW REMOVAL METHODS ON THE ISTD BENCHMARK

Model	RMSE ↓	RMSE-N ↓	RMSE-S ↓	SSIM ↑	SSIM-N ↑	SSIM-S ↑	PSNR ↑	PSNR-N ↑	PSNR-S ↑
Yang et al. [13]	15.63	14.83	19.82	0.901	0.912	0.929	23.01	23.35	29.33
Guo et al. [14]	9.3	7.46	18.95	0.919	0.944	0.978	23.07	24.86	30.98
Gong et al. [15]	8.53	7.29	14.98	0.908	0.929	0.98	24.07	25.26	32.43
DeShadowNet [16]	7.83	7.19	12.76	0.932	0.942	0.957	25.56	27.45	29.13
STC-GAN [1]	7.47	6.93	10.33	0.929	0.947	0.985	27.43	28.67	35.81
SID [17]	7.96	7.72	9.64	0.948	0.964	0.986	25.01	26.1	32.88
DSC [18]	6.67	6.39	9.22	0.845	0.885	0.967	26.62	28.18	33.45
RIS-GAN [19]	6.62	6.31	9.15	0.925	0.937	0.948	26.77	27.89	31.42
DHAN [20]	6.28	5.92	8.43	0.921	0.941	0.983	27.88	29.54	34.79
Auto-Exp [21]	5.88	5.51	7.9	0.845	0.879	0.975	27.19	28.6	34.71
G2R [22]	7.84	7.54	10.71	0.932	0.967	0.974	24.72	26.18	31.62
Mask-ShadowGAN [23]	7.63	7.03	10.35	0.921	0.943	0.953	27.44	28.54	30.11
LG-shadow [24]	6.67	5.93	11.51	0.906	0.938	0.974	25.83	28.32	31.08
SpA-Former [25]	6.86	6.22	10.48	0.931	0.956	0.982	27.73	30.16	33.51
ShadowRefiner [26]	9.12	6.31	7.32	0.857	0.881	0.952	26.04	28.64	30.54
ShadowFormer [3]	4.21	3.86	5.62	0.934	0.952	0.985	30.46	32.19	37.02
Diff-shadow [27]	7.12	5.43	6.43	0.901	0.945	0.950	29.43	32.00	35.13
MemoryNet	6.03	5.42	9.72	0.952	0.970	0.986	28.03	30.23	34.44



Fig. 4. Comparative visualization of raindrop removal results on the Raindrop benchmark.

despite its inclusion in their network inputs. Although Auto-exp [21] and G2R [22] effectively employ triplet datasets (input, mask, target) based on evaluative metrics, the need for shadow mask acquisition in real-world application settings is questionable. Our training only needs data pairs, while testing requires just a basic shadow image, offering significant practical benefits. DSC [54], and SpA-Former [25] may incorrectly handle relatively dark non-shaded regions bringing some misestimation. It turns out that their model fails to take full advantage of the shadow mask information, even if their network input contains a shadow mask.

2) *Rain Removal*: As presented in Table II and Fig. 4, we outline the PSNR/SSIM performance scores for rain removal

on the DeRainDrop testB and testA datasets. Our approach is evaluated against existing techniques, such as CMFNet [55], D-DAM [31], BPP [32], Maxim [34], IDT [35] and Rain-dropClarity [36]. On the test-b set, our MemoryNet attained the top SSIM score of 0.84 and the second-highest PSNR of 25.38 dB. In comparison, it achieved the highest SSIM score of 0.904 and the best PSNR of 24.64 dB on test-a. The visualization results for the DeRainDrop test-b image are illustrated in the figure. These findings indicate the superior effectiveness of our method in eliminating raindrops. The images produced by our technique more closely resemble the actual images than those produced by other models. Additionally, we carried out rain removal experiments using



Fig. 5. Qualitative comparison of motion deblurring performance on the GOPRO test set.

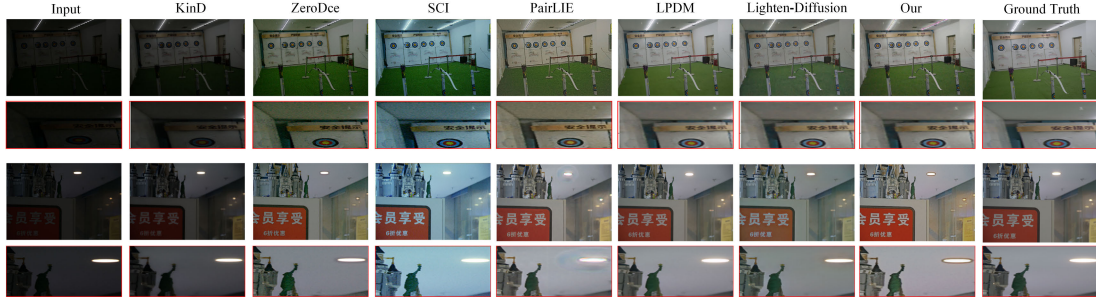


Fig. 6. Visual results comparing low-light enhancement performance using the LOL-v2 benchmark.

TABLE II
QUANTITATIVE RESULTS FOR RAINDROP REMOVAL
ON RAINDROP DATASET

Method	Venue	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
DeRaindrop [28]	CVPR2018	24.23	0.867	0.413
D-DAM [31]	Arxiv2021	23.71	0.892	0.421
BPP [32]	ICIP2021	23.93	0.883	0.432
MPRNet [33]	CVPR2021	23.12	0.891	0.201
MMOS [6]	CVPR2021	23.93	0.884	0.312
MAXIM [34]	CVPR2022	25.74	0.891	0.501
IDT [35]	PAMI2022	24.57	0.896	0.301
RaindropClarity [36]	ECCV2024	25.89	0.900	0.202
MemoryNet		24.64	0.904	0.231

TABLE III
MOTION DEBLURRING PERFORMANCE METRICS
ON THE GOPRO TEST SET

Method	Venue	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Gao et al. [37]	CVPR2019	29.67	0.928	0.421
DBGAN [38]	CVPR2020	29.87	0.935	0.410
MT-RNN [39]	ECCV2020	29.92	0.938	0.301
MPRNet [33]	CVPR2021	29.97	0.939	0.323
Restormer [40]	CVPR2021	32.92	0.961	0.331
DGUNet [41]	CVPR2022	31.48	0.951	0.331
MADANet [42]	CVPR2022	33.84	0.964	0.428
SFNet [43]	ICLR2023	33.27	0.963	0.331
Deblurdiff [44]	NeurIPS2025	34.99	0.969	0.232
MemoryNet		33.76	0.969	0.299

MMOS [6], a rain removal network that also incorporates a memory module. Nonetheless, MMOS yielded unrealistic results, proving ineffective on actual rain removal datasets. We believe this may be due to the unsuccessful integration of noisy data with pseudo-labels generated by the target network during real data processing.

TABLE IV
LOW-LIGHT ENHANCEMENT PERFORMANCE COMPARISON
ON THE LOL DATASET

Method	Venue	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
KinD [29]	ACM2019	17.65	0.775	0.271
ZeroDCE [45]	CVPR2020	14.86	0.595	0.207
RUAS [46]	CVPR2021	16.401	0.503	0.434
SCI [47]	CVPR2022	14.78	0.522	0.339
PairLie [48]	CVPR2023	19.51	0.736	0.248
GenerativePrior [49]	CVPR2023	12.55	0.412	0.401
NeRCO [50]	ICCV2023	19.70	0.771	0.321
CLIP-LIE [51]	ICCV2023	17.20	0.589	0.322
LPDM [52]	PR2024	19.20	0.793	0.322
Lighten-Diffusion [53]	ECCV2024	20.45	0.803	0.192
MemoryNet		19.85	0.826	0.228

3) *Image Deblurring*: Within Table III and Fig. 5, we present the PSNR/SSIM results for various methods applied to the deblurring task. Our analysis includes a comparison with numerous leading algorithms such as Gao [37], DBGAN [38], MT-RNN [39], MPRNet [33], DGUNet [41], MADANet [42], SFNet and DeblurDiff [43]. The results of the quantitative assessment are displayed in Table III. Despite MemoryNet not achieving the top scores, its performance remains commendable, illustrating its ability to handle image degradation effectively. DGUNet achieved superior performance metrics, which can be credited to its novel gradient strategy integrated into the proximal gradient descent (PGD) algorithm. This enables the model to address complex, real-world image degradation scenarios, thus enhancing its suitability for practical use.

4) *Low-Light Image Enhancement*: We compare our approach against KinD [29], Zero-DCE [44], SCI [46], PairLie [47], GenerativePrior [48], NeRCO [49], CLIP-LIE [50], LPDM [51] and Lighten-Diffusion [52]. The results of these comparisons are displayed in Fig. 6 and Table IV. KinD [29]

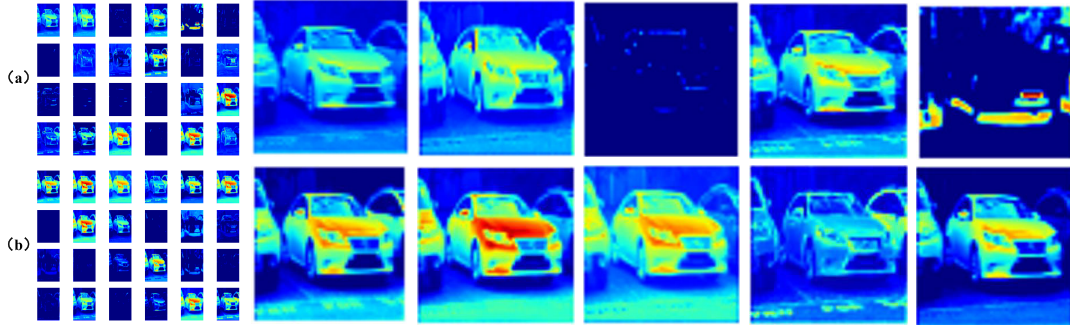


Fig. 7. Feature map visualizations comparing network activations: (a) baseline model without memory augmentation, (b) enhanced representations with memory augmentation applied.

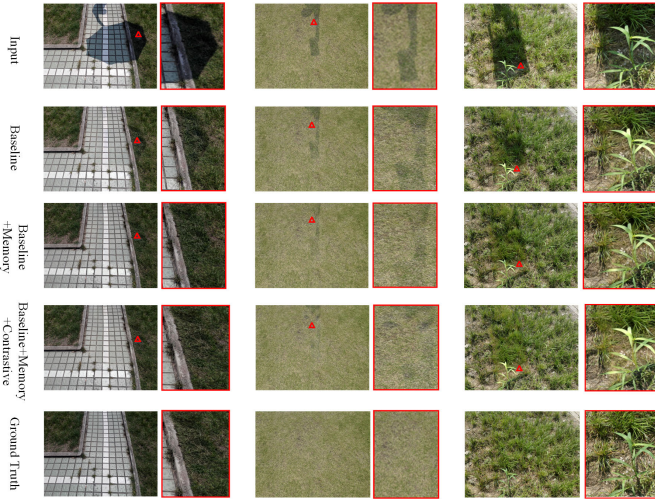


Fig. 8. Comparative feature visualizations from the initial layer: left shows raw features before processing, right displays enhanced features after memory augmentation.

enhances image details and contrast by decomposing images into illumination and reflection, effectively managing issues like overexposure and color distortion. However, this technique may not be ideal for images with highly uneven lighting conditions. Zero-DCE [44] employs a deep learning approach without a reference, adjusting well to diverse lighting, although it might result in unnatural color changes, compromising the image's natural appearance. SCI [46] introduces an innovative strategy for enhancing low-light images, offering rapid processing speed and adaptability. Despite impressive flexibility and speed, maintaining the image's naturalness and detail in low-light situations can be challenging. PairLie [47], LPDM [51] and Lighten-Diffusion [52] offer a technique for boosting low-light images by comparing them to their normal-light counterparts. The approach processes image pairs, which confines its usefulness to instances where a reference image is unavailable. Nonetheless, the method detailed in the paper yields acceptable outcomes concerning color and exposure, among other factors.

C. Ablation Study on MemoryNet

1) Quantitative Comparisons on Memory Augmentation:

In order to evaluate the efficacy of the Memory augmentation

TABLE V
COMPONENT ANALYSIS EVALUATING THE CONTRIBUTION OF MEMORY AUGMENTATION AND CONTRASTIVE LEARNING ON THE ISTD BENCHMARK

Baseline	Memory	Contrastive	RMSE	SSIM	PSNR
✓			6.74	0.921	26.79
✓	✓		6.26	0.932	27.80
✓		✓	6.21	0.951	27.68
✓	✓	✓	5.42	0.952	28.03

and contrastive learning techniques introduced in this study, we carried out ablation analyses on the ISTD datasets. These analyses are illustrated in Figs. 7 and 8, and Table V. Since each Memory item calculates cosine similarity with all queries, we explored different configurations of our Memory layer, including three-branch, two-branch, and single-branch models. The findings, as detailed in Table V, reveal that Memory augmentation is especially effective within a three-stage recovery network. Additionally, to clarify the influence of Memory augmentation, we visualized features within a compact network using a car image, as shown in Fig. 7. The comparative visualization indicates that the feature map with Memory augmentation is more consistent with the network's propagation, unlike the original feature map without augmentation, which diverges from the source image. The Memory augmentation network adeptly highlights the prominent features of the image, as demonstrated by the more distinct heatmap in the car image, confirming the successful identification of key features.

2) Quantitative Comparisons on Contrastive Learning:

Our study investigates contrastive learning as a substitute for generative learning. Unlike generative learning, contrastive learning emphasizes distinguishing differences between data on a conceptual level without examining the specific details of each example. This approach simplifies the model and optimization process, while also enhancing generalizability. We apply contrastive learning following the residual network, effectively converting it into a discriminator. This technique aids in crafting an encoder capable of producing similar representations for data in the same category, while amplifying the differences between representations of data from different categories. The outcomes of our research indicate that applying contrastive learning significantly enhances certain tasks, such

as de-shadowing. Notably, integrating a memory network with contrastive learning led to substantial improvements in metrics, with PSNR increasing by 1 point to 33.44, SSIM rising to 0.986, and RMSE decreasing to 6.03, as detailed in Table V.

V. CONCLUSION

This paper presents MemoryNet, a system that includes two key components: DA-CLIP and MemoryNet. DA-CLIP is tailored for perceptual classification of images with degradation, and MemoryNet features a tri-level memory layer and contrastive learning. Our tests across four different settings show the effectiveness of these approaches in improving image restoration outcomes. Additionally, the model put forth in this work exhibits significant improvements in both PSNR and SSIM metrics across three datasets with different degradation types, highlighting its capability to generate perceptually accurate restored images.

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